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PREPARED BY :		REVIEWED BY :	REVIEWED BY
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REVIEWED BY		APPROVED BY :	APPROVED BY :
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1. TUJUAN

PURPOSE

Memberikan petunjuk yang jelas mengenai cara perawatan terhadap Mesin *Filling Bosch FXS 2051* pada area produksi.

Provide clear instructions on how to maintenance for the FXS 2051 Filling Bosch Machine in the production area.

2. RUANG LINGKUP

SCOPE

SOP ini digunakan oleh Departement Engineering sebagai pedoman dalam melakukan perawatan mesin *Filling Bosch FXS 2051* pada area produksi.

This SOP is used by the Engineering department as a guideline for carrying out machine filling bosch FXS 2051 maintenance in the production area.

3. TANGGUNG JAWAB

RESPONSIBILITY

3.1. Teknisi Engineering *Preventive Maintenance* bertanggung jawab:

Engineering Technician Preventive Maintenance are responsible:

3.1.1. Melakukan tugas perawatan dengan baik, sesuai dengan SOP ini.

Performing maintenance duties properly, in accordance with this SOP.

3.1.2. Melakukan perawatan sesuai jadwal yang telah ditetapkan.

Perform maintenance according to a predefined schedule.

3.1.3. Melaporkan setiap terjadi kerusakan dan perbaikan yang diperlukan kepada Supervisor Engineering.

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Report any damage and repairs that are needed to the Engineering Supervisor.

3.2. Supervisor Engineering bertanggung jawab:

Engineering Supervisor are responsible:

3.2.1. Membuat jadwal perawatan secara berkala.

Make schedule of maintenance periodically.

3.2.2. Memastikan pelaksanaan perawatan sesuai dengan jadwal.

Ensure implementation of maintenance is according to the schedule.

3.2.3. Memastikan semua alat ukur yang terlibat dalam proses perawatan sesuai dengan spesifikasi yang telah ditentukan (dikalibrasi).

Ensure all measuring instruments involved in the maintenance process are in accordance with the specifications specified (calibrated).

3.2.4. Mengadakan tinjauan bulanan terhadap hasil kerja Teknisi Preventive Maintenance.

Perform monthly review of the workings of the Preventive Maintenance Technician.

3.2.5. Memberikan pelatihan kepada departemen lain yang terkait.

Provide training to others department which related.

3.3. Manajer Engineering bertanggung jawab:

Engineering Manager is responsible:

3.3.1. Menentukan spare part pengganti apabila spare part yang sesuai sudah tidak dijual lagi.

Determine replace parts if suitable spare part are no longer sold.

3.3.2. Memutuskan dan mengkoordinir tindakan yang diperlukan, apabila perlu dilakukannya penggantian, modifikasi peralatan, perubahan sistem dan menyiapkan prosedur perubahan.

Resolve and coordinate necessary actions, if necessary replacement, equipment modification, system changes and prepared change control.

3.4. Operator Mesin Produksi bertanggung Jawab untuk:

The Production Machine Operator is responsible for:

3.4.1. Melaporkan jika diareanya ditemukan kelainan-kelainan/kerusakan-kerusakan yang biasanya menghambat dalam pengoperasian mesin tersebut.

Report if the area is detected by defects / abnormalities that normally impede the operation of the machine.

3.4.2. Melakukan perawatan harian seperti : mengencangkan, klem-klem, ferulle, baut-baut yang kendur, membersihkan body mesin / Main Frame setiap saat sehingga tidak berkarat dan permukaan mesin tetap baik, membersihkan tiap-tiap mesin yang

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menjadi tanggung jawabnya, terutama pada bagian dalam dari kerak-kerak yang menempel.

Perform daily maintenance such as: tighten ferulle clamps loose bolts, clean the body of the machine / Main Frame at all times so that it does not rust and the surface of the machine remains good, cleaned every machine that is the responsibility, especially on the inside of the crust that is attached.

- 3.4.3. Memastikan kondisi alat-alat listrik dan elektronik yang merupakan bagian dari mesin yang ditempatkan pada daerah rawan air, agar selalu terhindar dari percikan air.

Ensure the condition of electrical and electronic equipment that is part of the machine placed in water-prone areas, in order to always avoid splashing water.

- 3.4.4. Melakukan pembuangan air kondensat pada *Air Service Regulator* pada mesin mesin yang dilengkapi dengan sistem pneumatik.

Conduct condensate water discharge at Air Service Regulator on machine equipped with pneumatic system.

- 3.5. Staff Kalibrasi dan Validasi dari Department QA bertanggung Jawab melakukan kalibrasi terhadap komponen komponen alat ukur yang digunakan, misalnya *Thermometer, Pressure Gauge, Thermo Controller* dll, dan melakukan kualifikasi secara rutin.

The Calibration and Validation Staff from Department QA is responsible for calibrating the components of the measuring instrument used, eg Thermometer, Pressure Gauge, Thermo Controller etc, and qualify regularly.

4. DEFINISI

DEFINITION

- 4.1. Mesin pengisian otomatis adalah mesin yang digunakan untuk membantu mengisi produk dalam bentuk cairan, pengisian produk cair dapat dilakukan dalam kemasan primer seperti botol, vial, ampul, maupun *prefilled syringe*. Penggunaan mesin *filling* cairan otomatis membuat volume dalam kemasan primer lebih akurat sesuai dengan volume yang sudah ditentukan.

Automatic filling machines are machines used to help fill products in liquid form, filling liquid products can be done in primary packaging such as bottles, vials, ampoules, or prefilled syringes. The use of automatic liquid filling machines makes the volume in the primary packaging more accurate in accordance with the specified volume.

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4.2. *Autonomous Maintenance* adalah perawatan mandiri mesin yang dilakukan oleh operator mesin. Bila selama ini operator hanya dilatih untuk mengoperasikan mesin, maka sudah saatnya untuk dilatih lebih lanjut.

Autonomous maintenance is an independent engine maintenance carried out by the machine operator. If all this time the operators are only trained to operate the engine, then it's time to be further trained.

5. PETUNJUK UMUM

GENERAL INSTRUCTION

5.1. Sebelum melaksanakan perawatan pencegahan, staf Engineering harus berkordinasi dengan operator terkait area tersebut untuk mempersiapkan areanya.

Before implementing preventive maintenance, engineering staff must coordinate with the operator about the area to prepare the area.

5.2. Pada saat pelaksanaan perawatan operator terkait harus mendampingi Staf Engineering untuk menghindari kesalahan pengoperasian saat mesin dijalankan atau operator bisa menjalani *Autonomous Maintenance* pada mesin tersebut.

When carrying out maintenance, the related operator must accompany Engineering staf to avoid operating errors when the machine is running or the operator can undergo autonomous maintenance on the machine.

5.3. Keselamatan kerja harus selalu diperhatikan pada saat perawatan pencegahan, gunakan APD (Alat pelindung diri) yang sesuai peruntukannya pada kondisi saat perawatan.

Work safety must be paid attention to at the time of preventive maintenance, use PPE (Personal protective equipment) which according to allotment in condition when maintenance.

5.4. Untuk penggantian komponen kritis atau *spare part* yang berdampak langsung dengan produk apabila diganti dengan beda merek dan tipe yang berbeda tetapi satu jenis maka dibuatkan pengendalian perubahan (*change control*).

For the replacement of critical components or spare parts that have a direct impact on the product when replaced with different brands and different types but one type, make change control.

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6. ALAT DAN BAHAN

TOOLS AND MATERIALS

6.1. ALAT

TOOLS

1 set tools box, pada saat melakukan perawatan pencegahan.

1 set of toolbox, when performing Preventive Maintenance.

6.2. BAHAN

MATERIAL

6.2.1. Ethanol 70%.

Ethanol 70%.

6.2.2. Kain Wipes.

Wipes Fabric.

6.2.3. Gowning

Gowning

7. PROSEDUR

PROCEDURE

Perawatan berkala setiap 1 bulanan (L1), 6 bulanan (L3) dan 1 tahunan (L4) pada mesin *Filling Bosch FXS 2051*.

Periodic maintenance every 1 month (L1), 6 month (L3) and 1 yearly (L4) of Filling Bosch FXS 2051 machine.

7.1. Perawatan 1 bulanan (L1).

Maintenance 1 month (L1).

7.1.1. Pemeriksaan Main Transport Belt.

Checking the main transport belt.



Gambar 1. Main Transport Belt.

Figure 1. Main Transport Belt.

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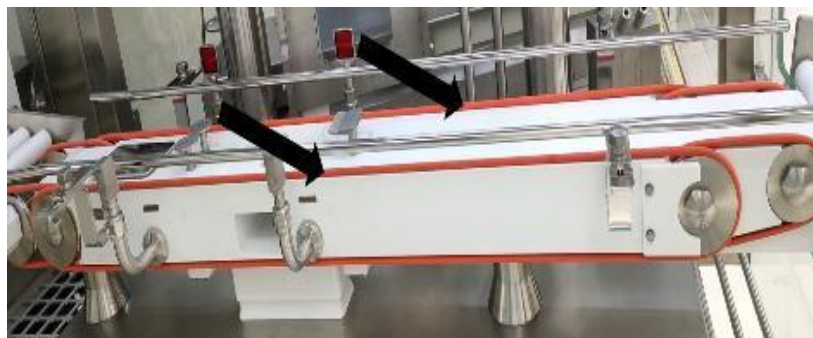
- 7.1.1.1. Periksa kondisi *Bearing Conveyor Infeed*, jika terdapat *Bearing* yang rusak ganti *Bearing*nya
Check condition bearing conveyor Infeed, if there are damaged bearings replace.



Gambar 2. *Bearing Conveyor Infeed*

Figure 2. *Bearing Conveyor Infeed*

- 7.1.1.2. Periksa kondisi *Round Belt*, kencangkan jika kendur dan ganti jika *Round Belt* terlihat rusak.
Check condition round belt, tighten if loose and replace if the round belt looks damaged.



Gambar 3. *Round belt*

Figure 3. *Round belt*

- 7.1.1.3. Periksa kondisi *Air Cylinder*, perbaiki jika tidak berfungsi baik atau ganti dengan yang baru.
Check the condition of the air cylinder, repair it if it doesn't work properly or replace it with a new.

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Gambar 4. Air Cylinder

Figure 4. Air Cylinder

- 7.1.1.4. Periksa kondisi sensor-sensor, jika ada yang tidak berfungsi dengan baik ganti sensornya, posisi sensor seperti pada gambar yang diberi tanda panah berwarna hitam

Check the condition of the sensors, if something doesn't work properly replace the sensor, sensors position as shown in the figure marked with a black arrow



Gambar 5. Sensor pada bagian Main Transport Belt

Figure 5. Sensor on the main part of the transport belt

- 7.1.1.5. Periksa bagian Bevel Gearnya, jika Gear terlihat kering beri pelumas Food Grade.

Check the bevel gear, if the gear looks dry give food grade lubricant.

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Gambar 6. Bevel gear

Figure 6. Bevel gear

- 7.1.1.6. Periksa kondisi baut L tanam *Pulley* pengikat as *Motor Conveyor*, kencangkan dan berikan *Locktike* jika baut L tanam kendur.

Check the condition of the L mounting bolts assembling conveyor pulleys, tighten them and provide a locktike if the L mounting bolts are loose.



Gambar 7. Pulley Mounting L Bolt

Figure 7. Pulley Mounting L Bolt

- 7.1.1.7. Periksa kondisi satu persatu *bearing-bearing* nya, jika terdapat *bearing* yang tersendat, bergoyang atau rusak ganti bearingnya

Check the conditions one by one the bearings, if there are rough, jammed or damaged bearings replace the bearing.

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Gambar 8. *The bearings*

Figure 8. *The bearings*

- 7.1.2. Pemeriksaan *Vacuum Pump*, posisi unit *Vacuum Pump* berada di area *CNC*.
Checking the Vacuum Pump, the position of the Vacuum Pump unit is in the CNC area.



Gambar 9. *Vacuum Pump*

Figure 9. *Vacuum Pump*

- 7.1.2.1. Bersihkan filter – filter *vacuum* pada bagian luar mesin maupun bagian dalam mesin, ganti jika filter sudah mulai rusak, contoh detail posisi filter pada gambar no. 10

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Clean the vacuum filter on the outside of the machine and the inside of the machine, replace if the filter has begun to break. Example of filter position details in figure no. 10.

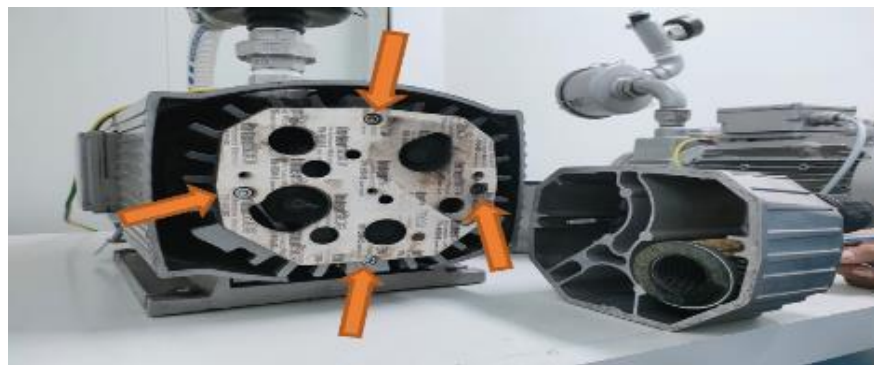


Gambar 10. Filter – filter Vakum

Figure 10. Vacuum filter

7.1.2.2. Kencangkan baut – baut pengikatnya.

Tighten the fastening bolts.



Gambar 11. Baut – baut

Figure 11. Bolts

7.1.3. Pemeriksaan *transport slide* dengan pelat tengah.

Transport slide check with centering plate.

7.1.3.1. Periksa kondisi *Flat Belt*, kencangkan jika kendur dan ganti jika *Flat Belt* terlihat rusak.

Check flat belt conditions, tighten if loosened and replace if the flat belt looks damaged.

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Gambar 12. Flat belt

Figure 12. Flat belt

- 7.1.3.2. Periksa kondisi *Timing Belt*, kencangkan jika kendur dan ganti jika *Timing Belt* terlihat rusak.

Check the timing belt condition, tighten if loose and replace if the timing belt looks damaged.



Gambar 13. Flat belt

Figure 13. Flat belt

- 7.1.3.3. Periksa kondisi satu persatu *Bearing*, jika terdapat *Bearing* yang tersendat, bergoyang atau rusak ganti *Bearingnya* dengan yang baru.

Check the conditions one by one the bearings, if there are rough jammed or damaged bearings replace the bearing with a new.

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Gambar 14. *Bearing*

Figure 14. *The bearings*

- 7.1.3.4. Periksa kondisi *Oil Seal* satu persatu, ganti jika terdapat *Oil Seal* yang rusak.

Check conditions oil seal one by one, replace if there is a broken oil seal.



Gambar 15. *Oil seal*

Figure 15. *Oil seal*

- 7.1.3.5. Periksa bagian *Pulley* dan *Gear*, lumasi dengan pelumas *Food Grade* jika bagian tersebut kering.

Check the pulley and gear parts, lubricate with food grade lubricant if the part is dry.

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Gambar 16. *Pulley dan gear*

Figure 16. *Pulley and gear*

- 7.1.3.6. Periksa kondisi *Slide Transport Rail* lumasi dengan pelumas *Food Grade* jika bagian tersebut kering.

Check the condition of the slide transport rail, lubricate with food grade lubricants if the parts are dry.



Gambar 17. *Slide transport rail*

Figure 17. *Slide transport rail*

- 7.1.3.7. Periksa keseluruhan baut-baut, jika ditemukan yang longgar ganti dengan yang baru.

Check all bolts, if you find the loose replace with a new.

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Gambar 18. Bolts

Figure 18. Bolts

7.1.4. Pemeriksaan *filling station*.

Checking station filling.

- 7.1.4.1. Periksa kondisi *Fitting* selang, jika terdapat yang rusak infokan kepada departemen produksi untuk segera diganti.

Check the condition of the hose fittings, if any are damaged, inform the production department to be replaced.



Gambar 19. *Fitting* selang

Figure 19. The hose fittings

- 7.1.4.2. Periksa kondisi setiap Servo Motor, jika Servo Motor tidak berfungsi dengan baik segera perbaiki atau ganti dengan yang baru dan infokan kepada bagian produksi untuk menentukan jadwal penggantianannya.

Check the condition of each servo motor, if the servo motor is not working properly, immediately repair or replace it with a new one and inform the production department to determine a replacement schedule.

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Gambar 20. Servo Motor

Figure 20. Servo Motor

- 7.1.4.3. Periksa kondisi *Linear Bearing* lumasi dengan *Grease Food Grade*, jika terdapat *Bearing* yang terlihat rusak ganti dengan yang baru.

Check the linear condition of the bearing for lubrication with food grade grease, if there is a bearing that looks damaged replace it with a new.



Gambar 21. Linear bearing

Figure 21. Linear bearing

- 7.1.4.4. Periksa kondisi *Bearing Cam Follower* lumasi dengan *Grease Food Grade*, jika terdapat *Bearing* yang terlihat rusak ganti dengan yang baru, dan bersihkan bagian *Cam Cam* nya lakukan *Greasing* ulang.

Check the condition of the cam follower bearing is lubricated with food grade grease, if there is a bearing that looks damaged replace it with a new one, and clean the cam cam part do greasing again.

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Gambar 22. *Bearing cam follower*

Figure 22. *Bearing cam follower*

- 7.1.4.5. Periksa kondisi *Level Sensor* pada *Buffer Tank*, bersihkan permukaan *Stick Sensor* dan periksa kabel koneksinya.

Check the sensor level conditions in the buffer tank, clean the sensor stick surface and check the connection cable.



Gambar 23. *Buffer Tank*

Figure 23. *Buffer Tank*

- 7.1.4.6. Periksa kondisi *Nozzle*, perbaiki jika *Nozzle* terlihat tidak *Center*, jika sulit untuk dilakukan perbaikan agar *Center* segera infokan kepada bagian produksi untuk dilakukan penggantian.

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Check the condition of the nozzle, repair it if the nozzle looks out of focus, if it is difficult to repair it, the center will immediately inform the production department for replacement.



Gambar 24. Nozzle

Figure 24. Nozzle

- 7.1.4.7. Periksa kondisi *Piston Pump* dan buka secara hati – hati, jika saat didorong *Manual Piston Pump* terasa kesat bersihkan menggunakan *WFI*, atau ganti *Seal Piston Pump* dengan yang baru jika sudah dilakukan cleaning masih terasa kesat.

Penggantian seal piston pump dilakukan ketika hour meter mencapai 100 dari data terakhir penggantian seal piston pump nya.

Check the condition of the piston pump and open it carefully, if when pushed manually the piston pump feels rough, clean it using WFI, or replace seal piston pump it with a new one if cleaning has been done, it still feels rough. Replacement of the piston pump seal is carried out when the hour meter reaches 100 from the last data on the replacement of the piston pump seal.

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Gambar 25. *Piston Pump*

Figure 25. *Piston Pump*

7.1.5. Pemeriksaan bagian bawah mesin.

Check the bottom of the engine.



Gambar 26. Bawah mesin

Figure 26. *The Bottom Of The Engine.*

7.1.5.1. Periksa kondisi *Bearing* dan beri pelumas *Food Grade*, jika terdapat *Bearing* yang rusak ganti dengan yang baru.

Check the condition of the bearings and give food grade lubricants, if there are damaged bearings replace with new.



Gambar 27. *Bearing Bearing*

Figure 27. *The Bearings*

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- 7.1.5.2. Periksa kondisi *Bearing Cam Follower*, lumasi dengan *Grease Food Grade*, jika terdapat *Bearing* yang terlihat rusak ganti dengan yang baru, dan bersihkan bagian *Cam* lakukan *Greasing* ulang.

Check the condition of cam follower bearing, lubricate it with food grade grease, if there is a bearing that looks damaged replace it with a new one, and clean the cam part, greasing it again.



Gambar 28. *Bearing Cam Follower*

Figure 28. *Bearing Cam Follower*

- 7.1.5.3. Periksa kondisi *Bearing Linear* dan beri pelumas *Food Grade*, jika terdapat bearing yang rusak ganti dengan yang baru.

Check bearing linear conditions and give food grade lubricants, if there are damaged bearings replace with new.



Gambar 29. *Bearing linear*

Figure 29. *Bearing linear*

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- 7.1.5.4. Periksa bagian *Timing Belt* dan *V-Belt*, jika terdapat bagian yang rusak ganti dengan yang baru.

Check the timing belt and v-belt, if there is a broken part replace it with a new.



Gambar 30. *Timing Belt* dan *V-Belt*

Figure 30. Timing Belt and V-Belt

- 7.1.6. Perawatan terhadap alat kontrol listrik dan panel *CIP SIP*.

Maintenance of electric control devices and SIP CIP panels.



Gambar 31. Alat kontrol listrik

Figure 31. Electric Control Devices

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7.1.6.1 Bersihkan panel kontrolnya menggunakan Vakum.

Clean the control panel using a vacuum.



Gambar 32. Panel kontrol

Figure 32. Control panel

7.1.6.2. Periksa koneksi – koneksi pada *Ferulle*, jika ditemukan terdapat bekas kebocoran pada sambungan *Ferulle* segera lakukan penggantian *Seal Ferulle* nya.

Check the connections on the ferulle, if traces of leakage are found at the ferulle connection, immediately replace the ferulle seal.



Gambar 33. *Ferulle*

Figure 33. Ferulle

7.1.6.3. Periksa semua kabel kontrol dan koneksi selang *Pneumatic*, pastikan terpasang kuat, kencangkan atau perbaiki jika terdapat yang kendur.

Check all control cables and pneumatic hose connections, make sure they are securely attached, tighten or repair any loose ones.

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Gambar 34. Kabel kontrol dan koneksi selang *Pneumatic Ferulle*

Figure 34. Control cables and pneumatic hose connections

- 7.1.6.4. Periksa kondisi semua tombol tekan darurat, ganti jika terdapat rusak.
Check the condition of all emergency push buttons, replace if there is damage.



Gambar 35. Tombol tekan darurat

Figure 35. Emergency push buttons

- 7.1.6.5. Periksa koneksi *Power* dan *Main switch HMI*, pastikan koneksinya tidak ada yg kendur segera perbaiki jika terdapat yg kendur.
Check the connection of the power and main switch of the HMI, make sure there is no loose connection immediately repair it if there is a loose connection.

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Gambar 36. Koneksi Power dan Main Switch HMI

Figure 36. Connection Power and Main Switch HMI

- 7.1.6.6. Periksa kondisi semua Valve GEMU Pneumatic System pada bagian panel CIP dan SIP, jika terdapat kebocoran pada seal ganti seal valve dengan yang baru.

Check the condition of all GEMU system pneumatic valve in the CIP and SIP panels, if there is a leak in the seal, replace the seal value with a new.



Gambar 37. Valve GEMU Pneumatic System

Figure 37. GEMU System Pneumatic Valve

- 7.1.6.7. Periksa kondisi filter Fan pada panel utama mesin, dan bersihkan filternya.

Check the condition of the fan filter on the main panel of the machine, and clean the filter.

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Gambar 38. Filter fan pada panel utama

Figure 38. Fan filter on the main panel

- 7.1.6.8. Bersihkan debu-debu yang menempel pada bagian elektrikal di dalam panel utama.

Clean the dust that sticks to the electrical parts inside the main panel.



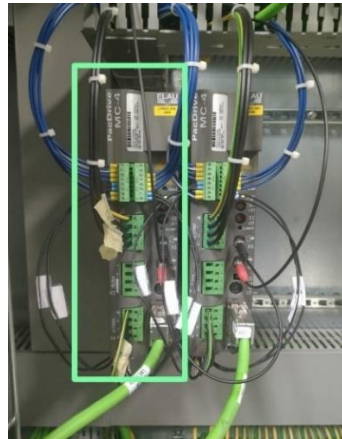
Gambar 39. Bagian electrical di dalam panel utama

Figure 39. Electrical parts inside the main panel.

- 7.1.6.9. Periksa lampu indikator pada Pacdrive MC4 dengan cara di nyalakan dahulu panelnya, indikator lampu normal adalah berkedip berwarna hijau, dan upnormal berwarna merah, segera laporkan kepada atasan jika lampu indikator berkedip berwarna merah.

Check the indicator light on the MC4 pacdrive by turning on the panel first, the normal indicator light is flashing green, and the abnormal light is red, immediately report to your superior if the indicator light is flashing red.

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Gambar 40. Pacdrive MC4

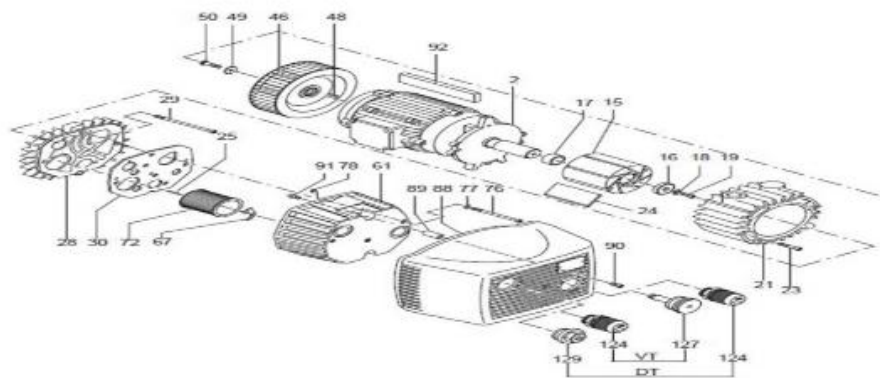
Figure 40. Pacdrive MC4

7.2. Perawatan 6 bulanan (L3).

Maintenance 6 month (L3).

7.2.1 Pemeriksaan Vacuum Pump.

Vacuum Pump Inspection.



Gambar 41. Vacuum Pump

Figure 41. Vacuum Pump

- 7.2.1.1 Periksa *Bearing Vacuum*, ganti *Bearing* jika putaran vakum terdengar kasar diputar secara manual sedikit kasar, contoh *Bearing* terdapat di nomer 16,17 dan bagian motor listriknya pada gambar No 37.

Check the vacuum bearing, replace the bearing if the vacuum rotation is rough. manually rotated a little rough, an example of the bearing is listed in number 16, 17 and the electric motor part in picture No. 37.

- 7.2.1.2. Periksa kipas pendinginan pada vakum, kencangkan jika kendur, contoh *Fan* pendingin terdapat di nomer 46 pada gambar No. 37.

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Check the cooling fan on the vacuum, tighten if loose, an example of a cooling fan is found at number 46 in figure No. 37.

- 7.2.1.3. Periksa sambungan pada AS motor listrik ke bagian *Housing Carbon Blade* kencangkan jika kendur atau ganti jika kondisinya terlihat kurang baik, contoh sambungan pada AS motor listrik ke bagian *Housing Carbon Blade* terdapat di nomer 15 pada gambar No. 37.

Check the connection on the AS of the electric motor to the carbon blade housing tighten if it is loose or replace if the condition looks bad, an example of the connection on the AS of the electric motor to the carbon blade housing is found at number 15 in figure No. 37.

- 7.2.1.4. Periksa kondisi *Carbon Bladenya*, ganti jika kondisi *Carbon Blade* terlihat rusak atau ukurannya berubah dari ukuran standarnya, ukuran standar menyesuaikan pada jarak *Housing Carbon Blade*, contoh *Carbon Blade* terdapat di nomer 24 pada gambar No. 37.

Check the condition of the carbon blade, replace if the condition of the carbon blade looks damaged or the size changes from the standard size, the standard size adjusts to the carbon blade housing distance, an example of a carbon blade is found at number 24 in figure No. 37.

- 7.2.1.5. Penggantian filter fan pada panel utama.

Replacing the fan filter on the main panel.



Gambar 42. *Fan Filter* pada panel utama

Figure 42. *Fan filter on the main panel*

- 7.3. Perawatan 1 tahunan (L4).

Maintenance 1 year (L4).

Penggantian filter udara model capsule 0,2µm untuk Point of use CA dan N2.

Replacement of 0.2µm capsule model air filters for Point of use CA and N2.

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7.4. Jika terjadi *Cold Start* pada mesin *Filling Bosch* segera hubungi tim Engineering lalu, setelah itu masuk ke *user* dan masukan *Passwordnya* dan lakukan *Reset* pada layar *Cold Start*, maka kondisi mesin akan kembali dalam keadaan normal, setelah itu lakukan percobaan pada mesinnya sebelum di gunakan kembali.

If a Cold Start occurs on the Bosch filling machine, immediately contact the Engineering team, then enter the user and enter the password and do a reset on the Cold Start screen, then the machine condition will return to normal, after that do a trial on the machine before using it again.

7.5. Lakukan pencatatan pada form Perawatan Berkala No. SOP/EBI/EN-024-F01 setelah selesai melakukan perawatan.

Record on the Periodic Maintenance form No. SOP/EBI/EN-037-F01 after finishing the maintenance.

8. REFERENSI

REFERENCE

Buku panduan perawatan berkala Mesin Pengisian Bosch FXS 2051.

Manual book of periodic maintenance Machine Filling Bosch FXS 2051.

9. LAMPIRAN

ATTACHMENT

Formulir No. SOP/EBI/EN-024-F01E Perawatan berkala terhadap Mesin Filling Bosch FXS 2051

Form No. SOP/EBI/EN-024-F01E Periodic maintenance of Bosch FXS 2051 filling machine.

10. RIWAYAT

HISTORY

Tanggal Date	No. Change Control Change Control No.	No. Revisi No. Revision	Alasan Perubahan Reason of Change
28 Dec 2021	CC-EBI-EN-017-21	01	Merubah level perawatan periodik yang pada awalnya setiap 3 bulan sekali (L2) dirubah menjadi 1 bulan sekali (L1). <i>Changing the level of periodic maintenance which was originally every 3 months (L2) changed to once a month (L1).</i>

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20 Dec 2022	CC-EBI-EN-066-22	02	7.2. Penambahan poin 6, mengenai alat dan bahan. <i>Addition of point 6, regarding tools and materials.</i> 7.3. Penambahan poin 7.2. mengenai tata cara Cold Start pada prosedur. <i>Addition of points 7.2. regarding Cold Start in the procedure.</i> 7.4. Penambahan poin 7.3. mengenai petunjuk pencatatan pada form Perawatan Berkala. <i>Addition of points 7.3. regarding the recording instructions on the Periodic Maintenance form.</i>
18 Jan 2024	CC-EBI-EN-016-23	03	1. Penambahan gambar spare part / mesin pada prosedur di setiap poin perawatan, agar memudahkan teknisi saat melakukan <i>Preventive Maintenance</i>. <i>Adding spare part / machine images to the procedure at each maintenance point, to make it easier for technicians when doing Preventive Maintenance.</i> 2. Penambahan perawatan 6 bulanan (L3) dan 1 tahunan (L4) pada prosedur, untuk poin pemeriksaan. <i>Addition of 6-monthly maintenance (L3) and 1 year (L4) to the procedure, for check points.</i>
	CC-EBI-EN-012-25	04	Penambahan poin 7.1.6.9 mengenai pemeriksaan lampu indikator pada pacdrive MC4 di prosedur. <i>Added point 7.1.6.9. regarding checking the indicator lights on the MC4 pacdrive in the procedure.</i>

11. DOKUMEN TERKAIT

RELATED DOCUMENT

Tidak ada

None

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12. TARGET PELATIHAN

TRAINING TARGET

Departemen Department	Jabatan Personil Personil Title
Departement Engineering <i>Engineering Departement</i>	Teknisi <i>Technician</i>

13. DISTRIBUSI

DISTRIBUTION

11.1. Departemen Engineering (*Hard copy*)

Engineering Department (Hard copy)

11.2. Departemen Sistem Mutu (QS) (*Hard copy*)

Quality System Department (Hard copy)

11.3. Departemen Produksi (*Hard copy*)

Production Department (Hard copy)

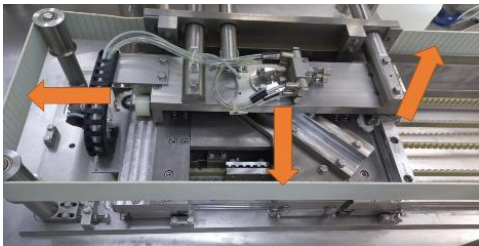
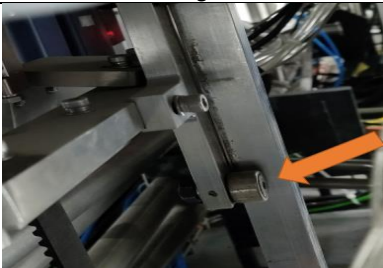
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Level :											
Month :											
Date :											
Day :											
Hours Meter :											
Level	Component	Activity							Condition		Remark
		C	I	L	R	S	T	W	B	G	
L1											
MAIN TRANSPORT BELT											
	Bearing conveyor infeed										
									Check condition infeed conveyor bearing, if there are damaged bearings replace		
	Condition Round Belt										
									Check condition belt round, tighten if loose and replace if the round belt looks damaged		
	Condition of the air cylinder										
									Check the condition of the air cylinder, repair it if it doesn't work properly or replace it with a new		
	Condition of the main transport belt										
									Check the condition of the sensors, if something doesn't work properly replace the sensor, sensors position as shown in the figure marked with a black arrow		

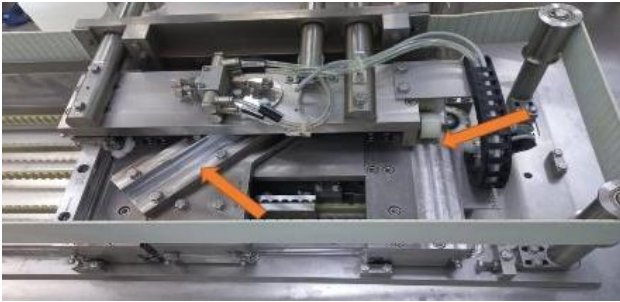
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Component	Activity							Condition		Remarks
	C	I	L	R	S	T	W	B	G	
Bevel gear										
							Check the bevel gear, if the gear looks dry give food grade lubricant			
Condition mounting L bolt										
							Check the condition of the L mounting bolts assembling conveyor pulleys, tighten them and provide a locktike if the L mounting bolts are loose			
Condition the bearings										
							Check the conditions one by one the bearings, if there are rough, jammed or damaged bearings replace the bearing			
VACUMP PUMP										
Condition Vacuum Pump										
							Checking the Vacuum Pump, the position of the Vacuum Pump unit is in the CNC area.			

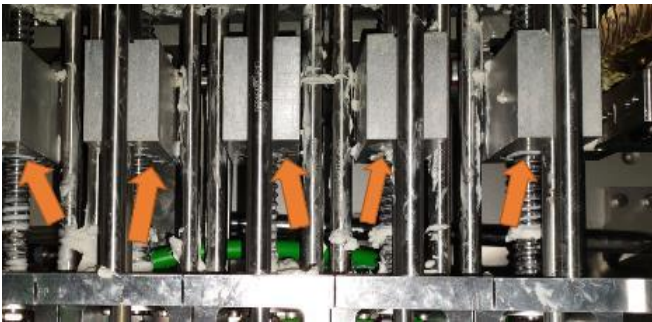
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	PERAWATAN BERKALA PADA MESIN FILLING BOSCH FXS 2051											
PERIODIC MAINTENANCE OF MACHINE FILLING BOSCH FXS 2051												

	Component	Activity							Conditoins		Remarks
		C	I	L	R	S	T	W	B	G	
	Condition Filter vacuum										
											Clean the vacuum filter on the outside of the machine and the inside of the machine, replace if the filter has ben broken.
	Condition Bolts										
											Tighten the fastening bolts
	TRANSPORT SLIDE CHECK WITH CENTERING PLATE										
	Condition flat belt										
											Check flat belt conditions, tighten if loosened and replace if the flat belt looks damaged.
	Condition flat belt										
											Check the timing belt condition, tighten if loose and replace if the timing belt looks damaged.
	Condition bearings										
											Check the conditions one by one the bearings, if there are rough jammed or damaged bearings replace the bearing with a new.

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	PERIODIC MAINTENANCE OF MACHINE FILLING BOSCH FXS 2051									

Component	Activity							Conditoins		Remarks
	C	I	L	R	S	T	W	B	G	
Condition Oil Seal										
							Check conditions oil seal one by one, replace if there is a broken oil seal			
Condition Pulley and Gear										
							Check the pulley and gear parts, lubricate with food grade lubricant if the part is dry			
Condition Transport Trail										
							Check the condition of the slide transport rail, lubricate with food grade lubricants if the parts are dry.			
Condition Bolts										
							Check all bolts, if you find the loose replace with a new.			

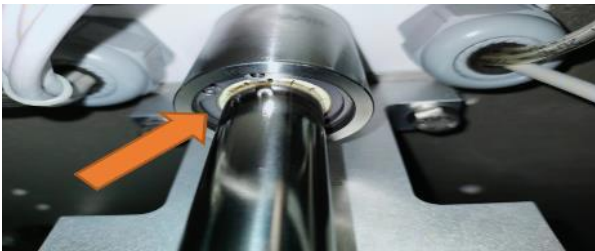
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Component	Activity							Conditioins		Remarks
	C	I	L	R	S	T	W	B	G	
CHECKING STASION FILLING										
Condition the hose fittings										
								Check the condition of the hose fittings, if any are damaged, inform the production department to be replaced.		
Condition Servo motor										
								Check the condition of each servo motor, if the servo motor is not working properly, immediately repair or replace it with a new one and inform the production department to determine a replacement schedule.		
Condition Linear bearing										
								Check the linear condition of the bearing for lubrication with food grade grease, if there is a bearing that looks damaged replace it with a new		
Bearing cam follower										
								Check the condition of the cam follower bearing is lubricated with food grade grease, if there is a bearing that looks damaged replace it with a new one, and clean the cam cam part do greasing again.		

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Component	Activity							Conditions		Remarks
	C	I	L	R	S	T	W	B	G	
Conditions buffer tank										
							Check the sensor level conditions in the buffer tank, clean the sensor stick surface and check the connection cable			
Conditions Nozzle										
							Check the condition of the nozzle, repair it if the nozzle looks out of focus, if it is difficult to repair it, the center will immediately inform the production department for replacement.			
Conditions Seal piston pump										
							Check the condition of the piston pump and open it carefully, if when pushed manually the piston pump feels rough, clean it using WFI, or replace seal piston pump it with a new one if cleaning has been done, it still feels rough. Replacement of the piston pump seal is carried out when the hour meter reaches 100 from the last data on the replacement of the piston pump seal.			

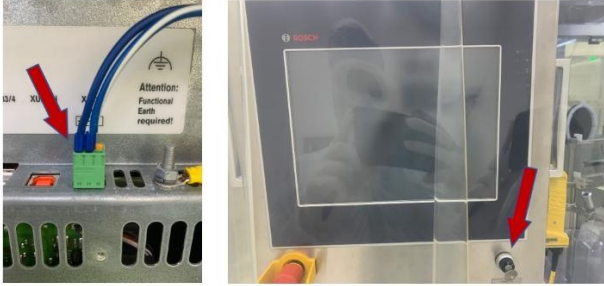
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PERIODIC MAINTENANCE OF MACHINE FILLING BOSCH FXS 2051										

Component	Activity							Conditions		Remarks
	C	I	L	R	S	T	W	B	G	
BOTTOM OF THE MACHINE										
Bearing										
								Check the condition of the bearings and give food grade lubricants, if there are damaged bearings replace with new.		
Bearing cam follower										
								Check the condition of cam follower bearing, lubricate it with food grade grease, if there is a bearing that looks damaged replace it with a new one, and clean the cam part, greasing it again.		
Linear bearing										
								Check linear bearing conditions and give food grade lubricants, if there are damaged bearings replace with new.		
Conditions Timming belt and V-Belt										
								Check the timing belt and v-belt, if there is a broken part replace it with a new.		

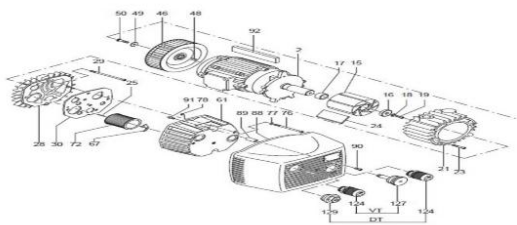
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Component	Activity							Conditions		Remarks
	C	I	L	R	S	T	W	B	G	
ELECTRIC & PANEL CIP SIP										
Conditions Control Panel										
								Clean the control panel using a vacuum.		
Conditions Ferulle										
								Check the connections on the ferulle, if traces of leakage are found at the ferulle connection, immediately replace the ferulle seal.		
Control cable and pneumatic connection										
								Check all control cables and pneumatic hose connections, make sure they are securely attached, tighten or repair any loose ones.		
Conditions Emergency pushbuttons										
								Check the condition of all emergency push buttons, replace if there is damage.		

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Component	Activity							Conditions		Remarks
	C	I	L	R	S	T	W	B	G	
Connection Power and Main Switch HMI										
							Check the connection of the power and main switch of the HMI, make sure there is no loose connection immediately repair it if there is a loose connection.			
Conditions Valve pneumatic system (GEMU)										
							Check the condition of all GEMU system pneumatic valve in the CIP and SIP panels, if there is a leak in the seal, replace the seal value with a new.			
Conditions Fan Filter on the main panel										
							Check the condition of the fan filter on the main panel of the machine, and clean the filter.			
Conditions Electrical Parts Inside the main panel										
							Clean the dust that sticks to the electrical parts inside the main panel.			

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	Component	Activity							Conditions		Remarks
		C	I	L	R	S	T	W	B	G	
	Condition Packdrive MC4										
								Check the indicator light on the MC4 packdrive by turning on the panel first, the normal indicator light is flashing green, and the abnormal light is red, immediately report to your superior if the indicator light is flashing red.			
L3	Vacuum pump										
											
	Check the bearing vacuum										
	Check cooling fan in the vacuum										
	Check connection in the AS vacuum										
	Check condition of the carbon brush										
	Fan filter in the main panel										
											
	Replacing the fan filter on the main panel.										
L4											
	Replacement of 0.2µm capsule model air filters for Point of use CA and N2										

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Note :								
No	Date	Start	Finish	Duration		Total personnel	Area	Remark
				Hour	Minute			
Remark : C : Cleaning (pembersihan) I : Inspection (pemeriksaan) L : Lubrication (pelumasan) R : Replacement (penggantian) S : Setting (penyetelan) T : Tightened (dikencangkan) W : Welding (pengelasan) G : Good (baik) B : Bad (jelek) * : Critical component			Level : L1 : Every 1 month L3 : Every 6 month L4 : Every 1 year		Done by : ()		Checked by : ()	

